

TraceX (SAHYOG) — Executive Q&A Quick Guide

SMART INDIA HACKATHON 2026 | PROBLEM STATEMENT: SIH26182 (MINISTRY OF HOME AFFAIRS / I4C)

Executive Purpose: A fast-track technical and operational reference guide designed for investigating officers, cyber cell analysts, and hackathon evaluators to understand the complete TraceX system in 5 minutes.

Section 1: Problem Statement & System Essentials

Q1: What is TraceX (SAHYOG) and what real-world crisis does it solve?

In modern cyber financial crimes (UPI fraud, digital arrest extortion, fake investment tasks, ransomware), perpetrators rapidly convert stolen Indian Rupees into cryptocurrency—primarily **USDT (TRC-20)** and **Bitcoin**—to evade domestic bank account freeze mechanisms.

- **The Operational Bottleneck:** Blockchain ledgers are pseudonymous. Investigating Officers (IOs) see only raw alphanumeric addresses (e.g. `TR7NHQJE...` or `0X3892...`) with zero visible names, phone numbers, or bank accounts. Tracing these manually across blockchain explorers takes weeks, allowing funds to escape.
- **The TraceX Solution:** TraceX is an autonomous crypto-forensic intelligence engine that traces illicit funds from the victim's wallet across complex multi-hop transaction trails to the terminal **Virtual Asset Service Provider (VASP/Exchange)** in **1 to 3 seconds**, generating court-admissible freeze notices in real time.

Q2: What is a VASP and why is it the pivotal target of crypto investigations?

VASP stands for **Virtual Asset Service Provider**—centralized cryptocurrency exchanges such as **CoinDCX, WazirX, Binance, ZebPay, and Mudrex**.

- **The Inevitable Cashout Chokepoint:** Scammers cannot spend USDT or Bitcoin to purchase groceries, vehicles, or real estate directly in India. To convert stolen crypto back into liquid INR cash, they must route funds to an exchange operating INR bank payouts or P2P trading desks.
- **KYC Identity Binding:** Regulated VASPs are mandated under Indian PMLA regulations to collect verified **Know Your Customer (KYC)** records: Aadhaar, PAN, phone numbers, linked bank accounts, and device IP access logs.
- **The Strategic Objective:** If law enforcement identifies the exchange's deposit address before fiat withdrawal occurs, the funds can be frozen under statutory order, and the perpetrator's real identity can be subpoenaed.

Section 2: End-to-End System Workflow (Part 1)

Q3: What is the step-by-step end-to-end workflow of TraceX?

TraceX operates through an automated 5-stage forensic pipeline:

Stage 1 Ingestion & Target Input: IO inputs suspect wallet address, FIR number, date of incident, and cyber crime portal reference ID.

Stage 2 Protocol & Chain Detection: Automated format and checksum verification (TRON Base58Check, Bitcoin Bech32/Base58, Ethereum EVM Hex).

Stage 3 Multi-Hop Graph Traversal: Breadth-First Search (BFS) explores transaction trails up to Hop 3+, distinguishing change outputs from pass-through mules.

Stage 4 VASP Attribution & AML Screening: Identifies destination exchange deposit clusters and evaluates FATF money laundering typologies to compute Risk Score.

Stage 5 Statutory Legal Dispatch & AI Guidance: Synthesizes Sec 106 BNSS Debit Freeze Directive, Sec 91 BNSS KYC Summons, and engages the Groq LPU Copilot for real-time guidance.

Section 2: End-to-End System Workflow (Part 2)

Q4: What are Multi-Hop Transaction Trails (Hop 0, Hop 1, Hop 2, Hop 3)?

Cyber syndicates route stolen funds through intermediate "mule" wallets to break direct attribution:

- **Hop 0 (Genesis / Inception):** The criminal's initial wallet where victim funds were first received.
- **Hop 1 (Primary Mule):** Layering stage where funds are partitioned into smaller amounts (peeling chain) to evade volume alerts.
- **Hop 2 (Secondary Mule / Aggregator):** Intermediate wallet recombining funds or bridging across blockchains.
- **Hop 3 (Terminal Gateway / Exchange Deposit):** The unhosted deposit address assigned to a user account at a centralized exchange (e.g. CoinDCX or Binance) where cashout takes place.

Q5: How is the VASP Attribution Confidence Score calculated?

Attribution is calculated using a deterministic mathematical formula:

$$\text{VASP Confidence Score} = \text{Match (+50\%)} + \text{Proximity (+25\%)} + \text{Regulation (+21\%)} - \text{Mixer Penalty (-30\%)}$$

- **Cluster Match (+50%):** Destination matches a verified hot wallet, sweeper, or deposit cluster of a known VASP.
- **Proximity Score (+25%):** Distance from Genesis: ≤ 2 hops = +25%; 3 to 4 hops = +15%.
- **Regulatory Alignment (+21%):** Identified VASP is registered as a Reporting Entity with FIU-IND (India) or global AML regulators.
- **Mixer Penalty (-30%):** Deducted if funds passed through Tornado Cash, ChipMixer, or privacy contracts.

Forensic Benchmark: Clean multi-hop paths to verified exchanges yield a **92% Confidence Score (High)**. Scores below 65% trigger "Manual Forensic Review Required".

Section 3: API Architecture & Ecosystem Breakdown

Q6: Which API does what in TraceX? (Complete Reference Matrix)

#	API / Service	Layer / Type	Core Forensic Role & Capability
1	TronGrid API	Blockchain RPC	Fetches real-time USDT (TRC-20) transfers, contract events, energy/bandwidth, and wallet balances on TRON.
2	Etherscan API v2	Blockchain Indexer	Tracks Ethereum & EVM transactions, ERC-20 stablecoin transfers, internal calls, and gas fee provenance.
3	Blockstream Esplora	Blockchain Indexer	Analyzes Bitcoin (BTC) UTXO inputs/outputs, mempool status, block confirmations, and address history.
4	Bitquery GraphQL v2	Cross-Chain Analytics	Traces decentralized exchange (DEX) liquidity pool swaps (Uniswap, PancakeSwap) and cross-chain bridge hops.
5	CoinGecko API	Market Intelligence	Provides real-time spot rates in USD and ₹ INR for statutory seizure valuation in Indian courts.
6	OFAC SDN Sanctions	AML & Sanctions DB	Screens wallets against US Treasury sanctioned entities, state-sponsored cyber groups (Lazarus), and illicit mixers.
7	ChainAbuse API	Threat Intel Feed	Ingests real-time crowd-sourced cyber threat reports, scam address complaints, and malicious contract telemetry.
8	Groq LPU (qwen3.8-27b)	AI Inference Hardware	Primary AI Copilot: Delivers sub-second (<1.0s) forensic intelligence and tactical guidance in English & Hindi.
9	Google Gemini 3.5 Flash	Cloud GenAI	Enterprise Cloud Failover: Deep multi-turn legal synthesis, complex typology reasoning, and resilient backup.
10	Neo4j Aura Cloud DB	Graph Database	Stores persistent entity graphs: Wallets modeled as Nodes and transfers as Directed Weighted Edges .

Section 4: Risk Score & FATF Typology Engine

Q7: What is the Risk Score and how is it calculated from 0 to 100?

The Risk Score is a quantitative probability of money laundering based on **FATF** and **Interpol** standards:

- **Base Score: 20 Points** (standard anomaly) or **35 Points** (wallet pre-flagged in threat databases).

FATF Typology Indicator	Points Added	Forensic Pattern & Behavioral Trigger
Peeling Chain	+20 Points	Sequential micro-transfers peeling off < 15% variation while forwarding the bulk balance to a new address.
Rapid Cashout	+25 Points	Inflow funds are forwarded to an exchange exit gateway within < 30 minutes of receipt.
Layering / Fan-Out	+30 Points	Funds from a single wallet are dispersed across ≥ 5 recipient mule accounts simultaneously.
Mixer / Privacy Protocol	+40 Points	Direct or 1-hop interaction with Tornado Cash, ChipMixer, or privacy-enhancing smart contracts.
Cross-Chain Bridge	+25 Points	Funds hop across independent blockchains (e.g. BTC → ETH → TRON) via bridge protocols.
Unhosted Transit Concentration	+15 Points	Transit path relies exclusively on private unhosted wallets lacking KYC identity verification.

Risk Tiers: **0-25: MINIMAL** **26-50: LOW** **51-70: MEDIUM** **71-85: HIGH** **86-100: CRITICAL**

Section 5: Statutory & Legal Framework (Indian Law)

Q8: What are Section 106 & Section 91 BNSS notices, and how is Section 65B BSA compliance achieved?

TraceX generates enforceable notices under India's new criminal justice framework:

- **Section 106 BNSS 2023 / Sec 102 CrPC (Debit Freeze Directive):**

Addressed to: Compliance Nodal Officer of the identified VASP (e.g. CoinDCX, Binance).

Statutory Power: Directs the exchange to execute an immediate electronic debit freeze on the destination deposit address and linked trading account, preventing the fraudster from withdrawing INR to bank accounts or executing P2P sales.

- **Section 91 BNSS 2023 / Sec 91 CrPC (KYC & Audit Logs Requisition):**

Addressed to: VASP Legal Directorate.

Statutory Power: Orders the exchange to furnish complete KYC dossiers (Aadhaar, PAN, phone number, bank account details, and login IP address logs with UTC timestamps) within 24 hours.

- **Section 65B BSA 2023 / Indian Evidence Act (Electronic Evidence Certificate):**

Every forensic report is cryptographically stamped with a deterministic **SHA-256 digital hash**, machine hardware UUID, system timestamp, and blockchain block header references, ensuring non-repudiation and strict admissibility in Indian trial courts.

Section 6: Dual-Engine AI Copilot Architecture

Q9: Why does TraceX utilize a Dual-Engine AI Copilot (Groq LPU + Gemini Flash)?

- Law enforcement operations require instant tactical answers—cloud LLM latency of 10 to 15 seconds is unacceptable when suspects are actively cashing out:
- **Primary Engine — Groq LPU (qwen3.8-27b):** Delivers inference in **800ms to 1.2 seconds** (sub-second response). Evaluates case telemetry and advises the officer in English or Hindi (e.g. *"Funds routed to CoinDCX deposit address. Click below to issue Section 106 Debit Freeze Directive."*).
 - **Enterprise Cloud Failover — Google Gemini 3.5 Flash:** Automatically takes over if the primary Groq endpoint experiences rate-limiting or network throttling, and conducts deep multi-document legal synthesis.
 - **Zero-Hallucination Guardrail:** The AI copilot is strictly bound to live on-chain JSON telemetry. It cannot hallucinate non-existent wallets, fabricate balances, or cite incorrect statutory sections.

Section 7: Two-Minute Winning Elevator Pitch for Judges

Q10: How should a team pitch TraceX to hackathon evaluators in 2 minutes?

Deliver these 4 core pillars with confidence:

- 1. The Problem:** "Judges, when a citizen loses ₹10 Lakhs in a cyber scam, fraudsters convert that fiat into crypto within minutes. Police investigations stall because blockchain wallets are pseudonymous strings of characters with zero names or phone numbers."

2. The Innovation: "Our system, **TraceX (SAHYOG)**, executes an automated Breadth-First Search graph traversal engine across Bitcoin, Ethereum, and TRON, linking that anonymous wallet to the exit exchange—like CoinDCX or Binance—in **just 1 to 3 seconds** with a **92% attribution confidence score**."

3. The Autonomous Action: "TraceX doesn't just draw graphs. It evaluates FATF money laundering typologies and instantly generates ready-to-dispatch **Section 106 BNSS Debit Freeze Directives** and **Section 91 BNSS KYC Summons** with Section 65B BSA cryptographic certification."

4. The Officer's Copilot: "Powered by a **Dual-Engine AI Copilot on Groq LPU and Gemini**, investigating officers get instant tactical intelligence in under 1 second. TraceX turns months of manual forensic backlog into real-time asset recovery."

EVALUATOR QUICK VERIFICATION SUMMARY (SIH 2026 CHECKLIST)

- ✓ **Multi-Hop BFS:** Tested up to 5 Hops

✓ **Real-time Speed:** 1 to 3 Seconds

✓ **VASP Match:** CoinDCX / Binance (92%)

✓ **Legal Compliance:** Sec 106/91 BNSS + 65B BSA